

2021*Time : 3 hours**Full Marks : 70*

*Candidates are required to give their answers
in their own words as far as practicable.*

The figures in the margin indicate full marks.

Answer all sections as directed.

Section-A**(Compulsory)**

1. Pick up the correct alternative for each of
the following questions : $10 \times 2 = 20$

(a) Full adder performs arithmetic sum of :

(i) Two binary input bits

☒ (ii) Three binary input bits

- (iii) Three input decimal numbers
- (iv) None of these
- (b) Choose which of the following is/are in POS form:
- (i) $(A' + C)(D' + C')$
- ☒ (ii) $A'C + D'C'$
- (iii) Both (i) & (ii)
- (iv) None of these
- (c) Don't care condition is denoted by X in K-map. What is the value for X ?
- (i) 0 and 1
- ☒ (ii) Either 0 or 1
- (iii) 1 and 2
- (iv) None of these
- (d) $F = \sum(1,3,4,5)$ is equivalent to :
- (i) $F = M_1 + M_3 + M_4 + M_5$

(ii) $F = m_1 + m_3 + m_4 + m_5$

(iii) $F = M_1 \cdot M_3 \cdot M_4 \cdot M_5$

(iv) $F = m_1 \cdot m_3 \cdot m_4 \cdot m_5$

(e) How many selection lines in 8 to 1 multiplexer ?

☒ (i) 3

(ii) 4

(iii) 2

(iv) 1

(f) If $J=1$ and $K=1$ in JK flip flop then the next state is :

(i) Predictable

(ii) Unpredictable

☒ (iii) Complement of previous state

(iv) None of these

(g) A gate, which is also known as inverter is :

(i) AND

(ii) OR

- ~~(iii)~~ NOT
- (iv) NAND
- (h) A register that keeps track of next instruction to be executed is called :
- (i) Program counter
- (ii) Instruction register
- (iii) Accumulator
- (iv) Data register
- (i) The fastest memory in a computer system is :
- (i) ROM
- (ii) RAM
- ~~(iii)~~ CACHE
- (iv) None of these

(j) ALU works on the instructions and data held in the :

(i) Notebook

☒ (ii) Registers

(iii) I/O devices

(iv) None of these

Section-B

Answer any **four** questions :

4×5=20

☒ 2. What is Half Adder ? Explain.

☒ 3. What is SR flip flop ? Explain.

4. Write short note on Indirect addressing mode.

☒ 5. Convert the following number system :

(a) $(12.8125)_{10} = ()_2$

(b) $(673)_8 = ()_{16}$

6. Draw the circuit diagram for the following boolean expressions :

(a) $AB' + A'B + A'B'$

(b) $(x + y') \cdot (x' + y) \cdot (x' + y')$

7. Define the following terminologies :

(a) Flip flop

(b) Logic gate

(c) Combinational circuit

(d) Multiplexer

(e) Register

Section-C

Answer any **two** questions of the following :

$$2 \times 15 = 30$$

8. Define universal gates. Realisation of NOT, OR, AND and NAND gates using NOR gate.

9. Prove the following boolean expressions by algebraic method :

$$(a) \quad (AB' + A'B)' = AB + A'B'$$

$$(b) \quad a'b' + bc + a'bc' = a' + bc$$

$$(c) \quad A'B(D' + C'D) + B(A + A'CD) = B$$

✓ 10. Simplify the following boolean function F together with the don't care condition d in :

(a) SOP form

(b) POS form $F = A'B'D' + A'CD + A'BC$

$$d = A'BC'D + ACD + AB'D'$$

✓ 11. Define Full Adder. Write truth table of Full Adder, derive sum and carry expressions from truth table and then design circuit diagram of Full Adder.

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